

Class Node & Class BrowserHistory Design

1. Class Node

1.1. General Description

The Node class represents a webpage that the user has visited. Each Node object is a link in the Doubly Linked List chain, storing webpage information while maintaining two-way links with adjacent Nodes.

[prev] <--- | url | title | timestamp | ---> [next]

1.2. Attributes

Attribute	Data Type	Description
url	String	Stores the URL of the webpage.
title	String	Stores the title of the webpage.
timestamp	long / Date	Stores the exact time the webpage was visited.
prev	Node	Pointer to the previously visited webpage.
next	Node	Pointer to the next webpage in the forward history.

1.3. Methods

Method	Description
Node(url, title)	Constructor to initialize a new Node with url, title, and current timestamp.
toString()	Returns a formatted string representing the Node for console printing.

2. Class BrowserHistory

2.1. General Description

The BrowserHistory class acts as the manager for the entire browser history, built on a Doubly Linked List data structure. This class uses three main pointers (head, tail, current) to track the state of the list and the user's current position.

head current tail
| | |

v v v
[Google] <=> [YouTube] <=> [GitHub] <=> [Facebook] <=> [Reddit]

2.2. Attributes

Attribute	Data Type	Description
head	Node	Pointer to the first webpage in the history.
tail	Node	Pointer to the last webpage in the history.
current	Node	Pointer to the webpage currently being viewed by the user.
size	int	Tracks the total number of webpages in the history.

2.3. Core Methods

Method	Description
visit(String url, String title)	Visits a new page. Inserts it after current. If branching occurs, truncates the forward history.
back()	Goes back to the previous page (current = current.prev).
forward()	Goes forward to the next page (current = current.next).
showHistory()	Traverses from head to tail and prints the entire history.

2.4. Helper Methods

Method	Description
canGoBack()	Returns true if the user can go back (current.prev != null). Used to disable UI buttons.
canGoForward()	Returns true if the user can go forward (current.next != null).
getCurrentPage()	Returns the current Node the user is viewing.
getSize()	Returns the total number of pages in the history.
isEmpty()	Returns true if the history is empty (head == null).
truncateForward()	(Private) Deletes the entire forward history branch when branching occurs. Severs links so the Garbage Collector can free memory.